

Scenario 3: Transfer Funds

Problem Statement

Customers should be able to transfer money from one account to another.

Before transferring funds:

1. Verify source account exists.
2. Verify destination account exists.
3. Check sufficient balance.
4. Deduct amount from source account.
5. Add amount to destination account.

Stored Procedure

```
CREATE OR REPLACE PROCEDURE TransferFunds(  
    p_from_account IN NUMBER,  
    p_to_account IN NUMBER,  
    p_amount IN NUMBER  
)  
AS  
    v_balance NUMBER;  
BEGIN  
  
    SELECT Balance  
    INTO v_balance  
    FROM Accounts  
    WHERE Account_ID = p_from_account;  
  
    IF v_balance >= p_amount THEN  
  
        UPDATE Accounts  
        SET Balance = Balance - p_amount  
        WHERE Account_ID = p_from_account;  
  
        UPDATE Accounts  
        SET Balance = Balance + p_amount  
        WHERE Account_ID = p_to_account;  
  
        COMMIT;  
  
        DBMS_OUTPUT.PUT_LINE('Transfer Successful');
```

```
ELSE
    DBMS_OUTPUT.PUT_LINE('Insufficient Balance');
END IF;

EXCEPTION

    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Account Not Found');

    WHEN OTHERS THEN
        ROLLBACK;
        DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);

END;
/
```

Explanation

- Accepts source account, destination account, and transfer amount as inputs.
- Retrieves the balance of the source account.
- Checks whether sufficient funds are available.
- Deducts the specified amount from the source account.
- Adds the same amount to the destination account.
- Displays a success message if the transfer is completed.
- Displays an error message if the balance is insufficient.
- Uses **ROLLBACK** if any error occurs.
- Saves successful transactions using **COMMIT**.